A. Extended Related Work and Comparison

A.1. LLM Prompting and CoT

Chain-of-Thought (CoT) prompting revolutionized how we elicit reasoning from Large Language Models by decomposing complex problems into intermediate steps (Wei et al., 2022). By augmenting few-shot exemplars with reasoning chains, CoT showed substantial performance gains on various tasks (Imani et al., 2023; Wei et al., 2022; Xu et al., 2024). Building on this, several variants emerged. Zero-shot CoT triggers reasoning without exemplars using instructional prompts (Kojima et al., 2022), and self-consistency enhances performance via majority voting over sampled chains (Wang et al., 2023). To reduce manual effort, Auto-CoT generates CoT exemplars using the models themselves (Zhang et al., 2023). Beyond linear chains, Tree-of-Thought (ToT) frames CoT as a tree search over partial reasoning paths (Yao et al., 2023), enabling lookahead and backtracking. SymbCoT combines symbolic reasoning with CoT by converting problems into formal representations (Xu et al., 2024). Recent work increasingly integrates CoT into the LLM inference process, generating long-form CoTs (Guo et al., 2025; Jaech et al., 2024; Team et al., 2025; Team, 2024). This enables flexible strategies like mistake correction, step decomposition, reflection, and alternative reasoning paths (Chen et al., 2025a; Yeo et al., 2025). The success of prompting techniques and long-form CoTs has led many to view them as evidence of emergent, human-like reasoning in LLMs. In this work, we investigate whether CoT reflects genuine reasoning or merely pattern interpolation.

A.2. Discussion on Illusion of LLM Reasoning

While Chain-of-Thought prompting has led to impressive gains on complex reasoning tasks, a growing body of work has started questioning the nature of these gains (Stechly et al., 2024). One major line of research highlights the fragility of CoT reasoning. Minor and semantically irrelevant perturbations such as distractor phrases or altered symbolic forms can cause significant performance drops in state-of-the-art models (Mirzadeh et al., 2025; Tang et al., 2023). Models often incorporate such irrelevant details into their reasoning, revealing a lack of sensitivity to salient information. Other studies show that models prioritize the surface form of reasoning over logical soundness; in some cases, longer but flawed reasoning paths yield better final answers than shorter, correct ones (Bentham et al., 2024). Similarly, performance does not scale with problem complexity as expected—models may overthink easy problems and give up on harder ones (Shojaee et al., 2025). Another critical concern is the faithfulness of the reasoning process. Intervention-based studies reveal that final answers often remain unchanged even when intermediate steps are falsified or omitted (Lanham et al., 2023), a phenomenon dubbed the illusion of transparency (Bentham et al., 2024; Chen et al., 2025b). Together, these findings suggest that LLMs are not principled reasoners but rather sophisticated simulators of reasoning-like text. However, a systematic understanding of why and when CoT reasoning succeeds or fails is still a mystery.

A.3. OOD Generalization of LLMs

Out-of-distribution (OOD) generalization, where test inputs differ from training data, remains a key challenge in machine learning, particularly for large language models (LLMs)(Budnikov et al., 2025; Yang et al., 2024, 2023; Zhang et al., 2024). Recent studies show that LLMs prompted to learn novel functions often revert to similar functions encountered during pretraining (Garg et al., 2022; Wang et al., 2024). Likewise, LLM generalization frequently depends on mapping new problems onto familiar compositional structures (Song et al., 2025). CoT prompting improves OOD generalization (Wei et al., 2022), with early work demonstrating length generalization for multi-step problems beyond training distributions (Shen et al., 2025; Yao et al., 2025). However, this ability is not inherent to CoT and heavily depends on model architecture and training setups. For instance, strong generalization in arithmetic tasks was achieved only when algorithmic structures were encoded into positional encodings (Cho et al., 2024). Similarly, finer-grained CoT demonstrations during training boost OOD performance, highlighting the importance of data granularity (Wang et al., 2025a). Theoretical and empirical evidence shows that CoT generalizes well only when test inputs share latent structures with training data; otherwise, performance declines sharply (Li et al., 2025; Wang et al., 2025b). Despite its promise, CoT still struggles with genuinely novel tasks or formats. In the light of these brilliant findings, we propose rethinking CoT reasoning through a data distribution lens: decomposing CoT into *task*, *length*, and *format* generalization, and systematically investigating through controlled experiments.

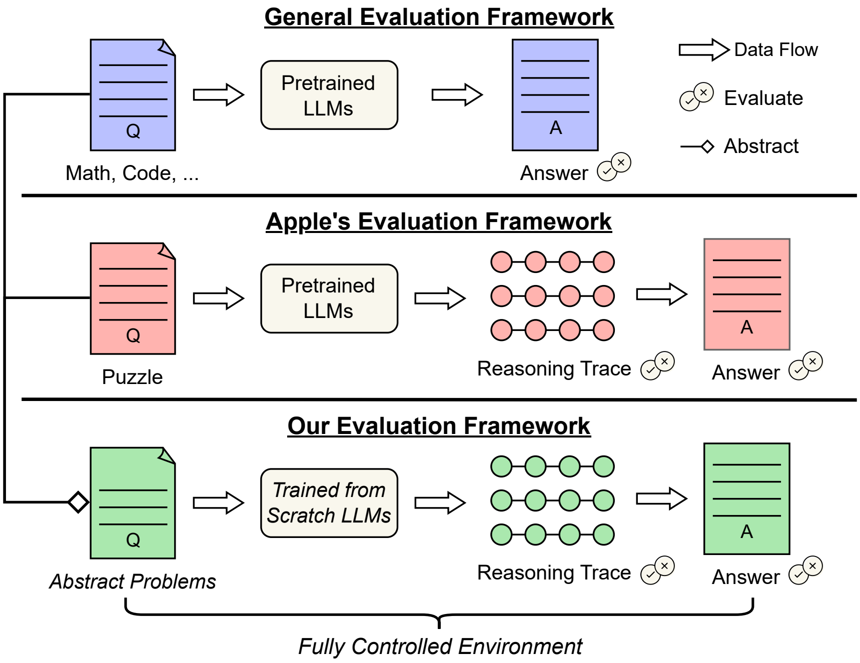

Figure 10 | Comparison with representative evaluation method. DataAlchemy distills real-world NLP problems, allowing training LLMs from scratch to avoid data leakage issues and study CoT reasoning through rigorous controlled experiments.

A.4. Comparison with Representative Work

Recent research attempts to create a fine-grained evaluation by examining both the CoT reasoning trace and the final answer through controlled experiments. However, they suffer from: (*i*) Narrowly defined settings: focusing on specific tasks, domains, or LLMs, thereby overlooking the common characteristics across tasks and LLMs. (*ii*) Data entanglement: most evaluations are conducted on real-world tasks and models, where the complexity precludes fully controlled experiments. (iii) Data leakage: LLMs train makes the best of all the data, including benchmarks, undermining the effectiveness and validity of evaluations, which is illustrated in Figure 10.

To scientifically and rigorously examine CoT reasoning, a new evaluation framework is required. An ideal framework should satisfy the following criteria: (*i*) Abstract representation: it should abstract and unify diverse NLP tasks and LLMs while retaining their essential properties. (*ii*) Fully controlled experiment: it should enable a full and fine-grained control of both tasks (e.g., complexity) and LLMs (e.g., size and architecture), enabling rigorous study of different factors through controlled